INCH-POUND

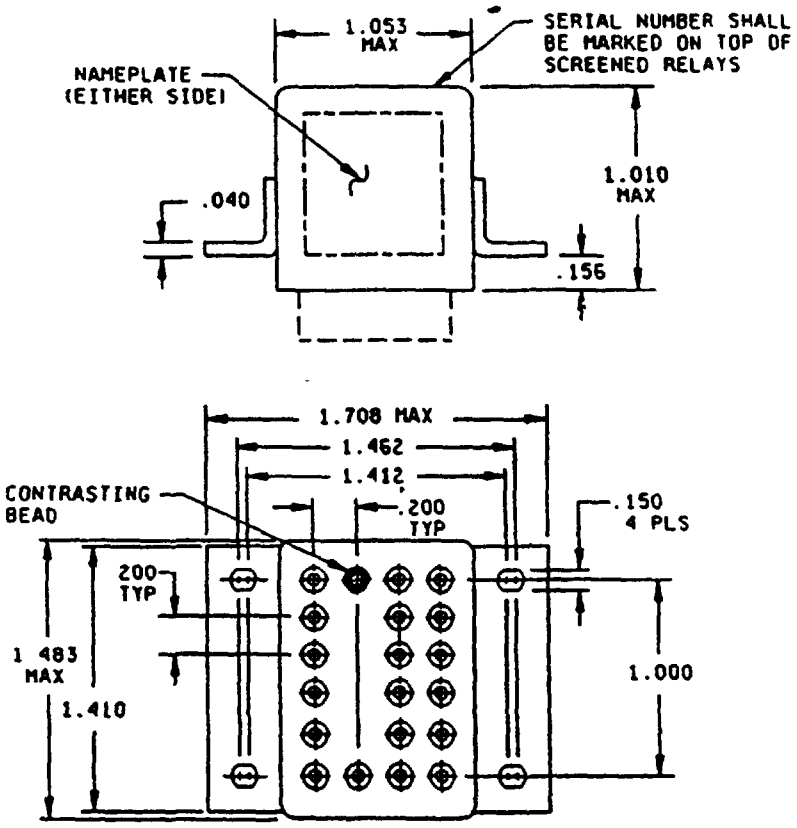
MIL-R-6106/8D
2 July 1993
SUPERSEDING
MIL-R-6106/8C
4 June 1985

MILITARY SPECIFICATION SHEET

Ⓓ RELAYS, ELECTROMAGNETIC, TYPE I, PERMANENT MAGNET DRIVE, 10 AMPERES, 6 PDT, HERMETICALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the relay described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-R-6106



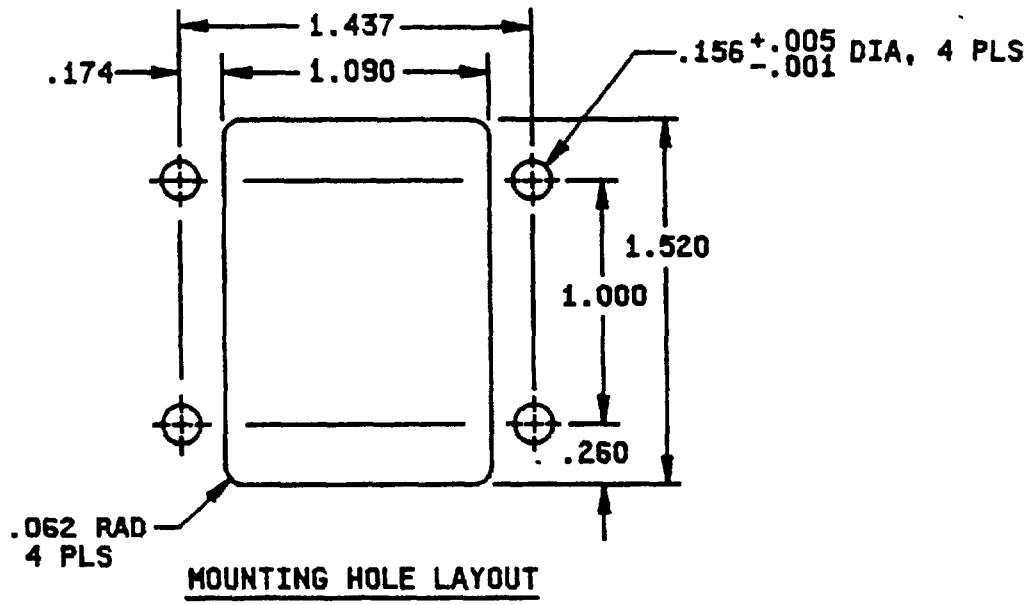
TERMINAL LAYOUT FOR DASH NUMBERS -001,-002,-003,-005,-006,-007,-009,-010,-011,-012, -013 AND -014 (SOLDER HOOK AND DC COIL PLUG-IN)

FIGURE 1 Dimensions, configuration, mountings, and circuits

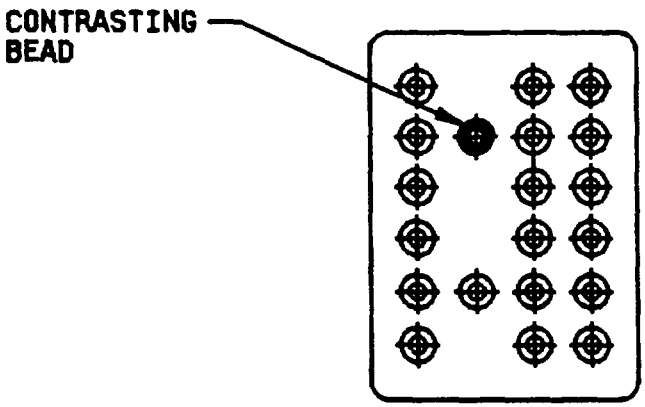
Ⓓ denotes changes

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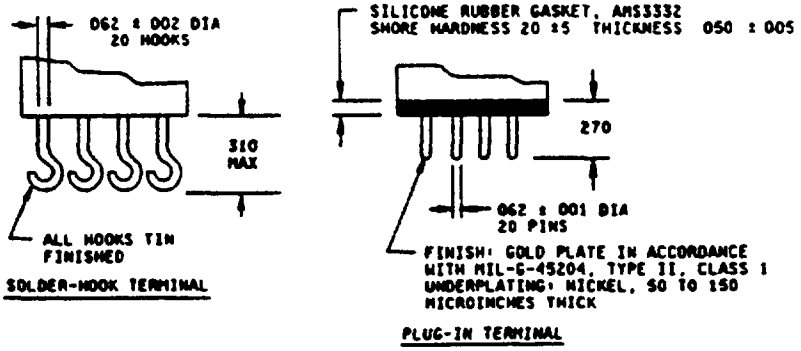
MOUNTING HOLE LAYOUT

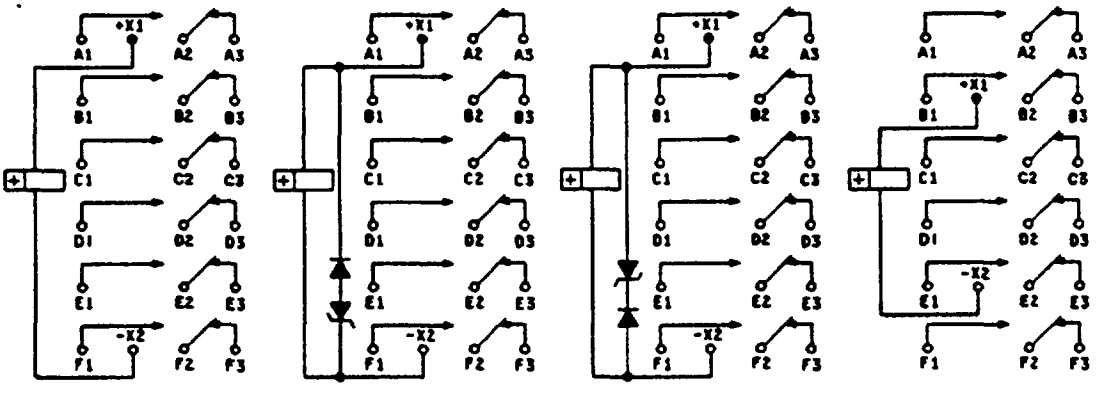


TERMINAL LAYOUT FOR DASH NUMBERS
-004 AND -008 (AC COIL PLUG-IN)

FIGURE 1 Dimensions, configuration, mounting, and circuits - Continued

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	0.03	.150	3.81	.310	7.87	1.412	35.86
.002	0.05	.156	3.96	1.000	25.40	1.437	36.50
.005	0.13	.174	4.42	1.010	25.65	1.462	37.13
.040	1.02	.200	5.08	1.053	26.75	1.483	37.67
.050	1.27	.260	6.60	1.090	27.69	1.520	38.61
.062	1.57	.270	6.86	1.410	35.81	1.708	43.38

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 inch (0.25 mm).
4. Terminal numbers shall not appear on the relay header. There shall be affixed to the relay a legible circuit diagram that identifies each terminal location specified.
5. Coil polarity is not applicable to ac versions of -003, -004, -007, and -008.
6. DC versions of this relay shall not operate or be damaged by reverse polarity. Semiconductors shall not be used for this purpose.
7. Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electromagnetic flux.

FIGURE 1. Dimensions, configuration, mounting, and circuits - Continued.

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REQUIREMENTS

CONTACT DATA

Life Unless otherwise specified 100,000 cycles

Load ratings

High level (relay case grounded)

Resistive 10 amperes at 28 V dc, 115 V ac, 400 Hz, 1 phase; 115/200 V ac, 400 Hz, 3 phase

(D) Inductive: 8 amperes at 28 V dc; 115 V ac, 400 Hz, 1 phase; and 115/200 V ac, 400 Hz, 3 phase (life: 20,000 cycles) For -003, -004, -007, and -008 inductive load shall be 8 amperes at 28 V dc for 10,000 cycles

(D) Motor 4 amperes at 28 V dc; 3 amperes at 115 V ac, 400 Hz, 1 phase and 115/200 V ac, 400 Hz, 3 phase.

(D) Lamp 2 amperes at 28 V dc and 115 V ac, 400 Hz, 1 phase

Intermediate current. Applicable.

Transfer load: Not applicable.

(D) Mechanical life (reduced current): 2 5 amperes at 28 V dc; 115 V ac, 400 Hz, 1 phase; and 115/200 V ac, 400 Hz, 3 phase.

Contact voltage drop or resistance·

High level·

Initial: 0.150 volt maximum.

After life: 0.175 volt maximum.

Contact bounce. 1.0 millisecond maximum.

Break bounce (normally open contacts only)· 0 1 millisecond maximum (applicable to dc suppressed coils and ac coils only).

Overload current· 40 amperes dc; 60 amperes ac

Rupture current: 50 amperes dc; 80 amperes ac

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Time current relay characteristics Applicable (see table I)

TABLE I Time current relay characteristics 1/ 2/

1	15 A	1 hour
2	50 A	5.0 seconds
3	100 A	1 2 seconds
4	250 A	0.2 second
5	350 A	0.1 second

1/ CAUTION: Compare with time current characteristics of the associated circuit protection device

2/ All relays shall withstand overload and fault currents Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of four different current levels in the sequence listed in table I. Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz, 3 phase. Cooling time between successive applications shall be 30 minutes. Tests shall be performed on both normally open and normally closed contacts There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

COIL DATA: See table II. CAUTION Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations

a. Row to row assisting fields. .125 inch.

b Row to row opposing fields 1875 inch.

c Side to side alternating fields. 0625 inch

d. Side to side like fields: .125 inch.

Operate time (dc coils): 15 milliseconds maximum with rated coil voltage.

Operate time (ac coils): 20 milliseconds maximum with rated coil voltage.

Release time (dc coils). 15 milliseconds maximum from rated coil voltage.

Release time (ac coils): 50 milliseconds maximum from rated coil voltage.

Duty rating Continuous.

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Back EMF (coil transient): 42 V dc maximum (applicable to -009, -010, -011, and -012).

Test temperature: Room ambient.

Point of measurement: Across coil terminals X_1 - X_2.

Coil voltage: 28 V dc.

Test method: Apply coil voltage to X_1 and X_2. With oscilloscope monitoring the coil terminals, measure and record the back electromotive force when voltage is removed.

ELECTRICAL DATA:

(D) Insulation resistance (initial): 100 megohms, minimum.

(D) Insulation resistance (after life or environmental tests): 50 megohms, minimum.

Dielectric withstanding voltage (sea level):

	Initial V rms 60 Hz	After life tests V rms 60 Hz
Coil to case	1,000	750
All other points	1,250	1,000

Dielectric withstanding voltage (altitude):

	80,000 ft V rms 60 Hz	300,000 feet V rms 60 Hz
Coil to case	350	500
All other points	350	500

NOTE: Dielectric may be improved by suitable insulation of terminals and wiring after installation.

ENVIRONMENTAL CHARACTERISTICS:

Operating temperature range: -70°C to +125°C.

Maximum altitude rating: 300,000 feet.

Shock (specified pulse): MIL-STD-202, method 213, test condition C, except peak value shall be 50 g's with a duration of 6 to 9 milliseconds. Contact chatter shall not exceed 10 microseconds for closed contacts and 1 microsecond maximum for open contacts.

(D) Vibration (sinusoidal): Applicable: 20 g's maximum with frequency range of 10 to 2,000 Hz. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum for open contacts. NOTE: Vibration test curves as shown in MIL-STD-202, method 204, figure 204-1 apply.

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(D) Vibration (random): MIL-STD-202, method 214, test condition 1F, 15 minutes each plane.

Acceleration: Applicable (15 g's maximum).

PHYSICAL DATA:

Dimensions, configuration, mounting, and circuits: See figure 1.

Terminal solderability: Applicable only to solder hook terminals.

(D) Weight (maximum): 4.16 ounces (117.94 grams).

Seal: Hermetic relays shall be sealed by welding, i.e., laser, TIG (tungsten inert gas), or other means approved by the qualifying activity.

Construction (internal and external): All welded, except that coil magnet wire to coil lead wire shall be soldered.

Identification of product: Applicable. The circuit diagram shall be oriented so that when the relay is held with the circuit diagram right side up as shown on figure 1, then rotated away from the viewer about a horizontal axis through the diagram until the header terminals face the viewer, each terminal shall be in the location shown on the circuit diagram.

Part or Identifying Number (PIN): M6106/8- (plus dash number from table II). PIN example: M6106/8-003.

TABLE II. Dash numbers and operating characteristics.

PIN M6106/8	Coil data										Terminals	Circuit schematic
	Coil	Rated		Max		Max pick-up voltage			Drop-out voltage 2/	Hold voltage 2/		
		Volts 1/	Freq	Volts	A	Normal 2/	High temp test	Cont current test				
-003	AC	115	400	122	0.06	90	95.4	103.5	5.0	30.0	Solder	A
-004	AC	115	400	122	0.06	90	95.4	103.5	5.0	30.0	Plug-in	C (D)
-005 3/	DC	28	N/A	29	0.18	18	19.8	22.5	1.5	7.0	Solder	A
-006 3/	DC	28	N/A	29	0.18	18	19.8	22.5	1.5	7.0	Plug-in	A
-007 3/	AC 4/	115	400	122	0.06	90	95.4	103.5	5.0	30.0	Solder	A
-008 3/	AC 4/	115	400	122	0.06	90	95.4	103.5	5.0	30.0	Plug-in	C (D)
-009	DC	28	N/A	29	0.18	18	19.8	22.5	1.5	7.0	Solder	B 4/
-010	DC	28	N/A	29	0.18	18	19.8	22.5	1.5	7.0	Plug-in	B 4/
-011 3/	DC 4/	28	N/A	29	0.18	18	19.8	22.5	1.5	7.0	Solder	B 4/
-012 3/	DC 4/	28	N/A	29	0.18	18	19.8	22.5	1.5	7.0	Plug-in	B 4/
-013	DC	28	N/A	29	0.18	18	19.8	22.5	1.5	7.0	Solder	A
-014	DC	28	N/A	29	0.18	18	19.8	22.5	1.5	7.0	Plug-in	A

1/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ Type I ER screened relays shall be tested to the group A type I, ER requirements only. Additional life testing is not required. For group A inspection, operational reliability shall be performed at the alternate low level run in.
4/ When semiconductors are required, JAN TX or equivalent screened semiconductors shall be used. Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode. Diodes shall have a peak inverse voltage of 600 V dc min.

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Qualification by similarity See table III

TABLE III Qualification by similarity

PIN M6106/8-	Loads		Dynamics 1/		Environmental	
	2/ A	2/ B	2/ A	2/ B	A	B
001	4		2		4	
002	4		2		4	
003		2,3		1		2,3
004		2,3		1		2,3
005	4		2		4	
006	4		2		4	
007		2,3		1		2,3
008		2,3		1		2,3
009	2,3		1		2,3	
010	2,3		1		2,3	
011	2,3		1		2,3	
012	2,3		1		2,3	
013	4		2		4	
014	4		2		4	

1/ All units must be tested. See dynamics section of MIL-R-6106, appendix.

2/ Terminals Socket pin type terminals are considered worse case Solder hooks may be qualified by similarity

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GROUP C INSPECTION

Group C inspection See table IV.

TABLE IV. Group C inspection

Coils	Number of samples
DC	2
DC suppressed	1
AC 1/	1

1/ If the manufacturer does not choose to retain qualification for the ac coils, substitute a dc coil for the ac coil

SUPERSESSION DATA

Supersession data See table V

TABLE V Supersession data

Superseded PIN M6106/8	New PIN M6106/8
-001	-013
-002	-014

CONCLUDING MATERIAL

Custodians.
Navy - EC
Air Force - 85

Review activity.
Air Force - 99

Preparing activity
Air Force - 85

Agent:
DLA - ES

(Project 5945-0882)

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